

Bishal Dhungana

Geospatial Data Scientist | AI/ML | Cloud Platform Engineering

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SUMMARY

Geospatial Data Scientist working across applied AI, cloud data platform engineering, and hazard-resilient infrastructure. Technical lead and acting product owner on FPL's AI-based infrastructure inspection program — independently built and deployed a computer vision model to production (AWS SageMaker, ECR), defined the ML scope and data pipelines behind a multi-vendor AI engagement, and built the GIS applications used for real-time disaster response. Background in field geomatics and peer-reviewed flood-hazard research. Open to Data Science, GIS/Geospatial Engineering, and AI/ML roles.

SKILLS

Machine Learning & Computer Vision: PyTorch, TensorFlow/Keras, Ultralytics, YOLO, RT-DETR, Detectron2, ResNet, XGBoost, OpenCV, ONNX

Generative AI & Agentic Systems: LangChain, LangGraph, RAG, Model Context Protocol, FastMCP, Vector Databases, Ollama, Chroma, Hugging Face Transformers

Data Engineering: Python, SQL, ETL/ELT Pipelines, DuckDB, Databricks, Data Lakes, Parquet

Geospatial: ArcGIS Pro / Enterprise / Online, Utility Network, PostGIS, DuckDB Spatial, GDAL, PDAL, GeoPandas, Shapely, PyProj, LiDAR / Point Cloud, Overture Maps, Cloud-Optimized GeoTIFF, Folium, FME

Cloud & DevOps: AWS Lambda, S3, SageMaker, ECR / ECS, AWS CDK, Docker, IAM, VPC / Network Security, CI/CD

Software Engineering: JavaScript / TypeScript, R, FastAPI, React, Streamlit

Field: GNSS / GPS Survey, UAV Survey (FAA Part 107), Differential GNSS, Geodatabase Design

EXPERIENCE

Data Scientist — Geospatial & AI — Florida Power & Light - Contract

Apr 2025 – Present

- Defined ML scope and technical specs as data scientist/acting product owner for a multi-million-dollar AI vendor engagement; authored RFP requirements and served as technical SME through development and benchmarking.
- Independently built and deployed an overhead switch-state (open/closed) classification model — trained locally, then migrated to AWS (ECR, SageMaker) for production inference.
- Evaluated CV architectures (YOLO, Detectron2, RT-DETR) using a precision/recall/F1 and image-scale methodology I designed; selected RT-DETR for transfer learning and standardized the approach for vendors.
- Built an API-based metadata pipeline with spatial (10m/30ft) and temporal (30-day) joins matching 350K+ condition reports to an 8M-image catalog, enriching ~2M images with condition metadata.
- Led the pipeline's move from ad hoc Python/GeoJSON scripts to an FME-to-PostGIS pipeline — designed the schema, migrated v1 to v2, and cut 100+ metadata columns to ~50 with spatial indexing and Azure-compatible storage.
- Designed the annotation-storage schema for multiple model versions, and built the taxonomy's operational layer — feature layers, REST APIs, front-end UI, audit logging — replacing a fragile Excel-based process.
- Built a Streamlit app (later migrated to Power BI) with REST APIs summarizing dataset availability and taxonomy patterns for leadership.

- Acted as product owner for a dataset-creation plugin (with the enterprise IT team) — defined filters, UI, and delivery logistics — used to deliver 300+ curated imagery datasets to vendors.
- Evaluated ~20 vendor bids with a GPT-based scoring agent, then audited 3 external ML vendors post-award for licensing compliance, resolving CUDA/GPU and batch-processing mismatches against AWS SageMaker.
- Co-developed an equipment detection and segmentation model (with a data scientist and software engineer) spanning 27-30 asset classes, evolving from object detection to segmentation.
- Designed an AWS CDK infrastructure-as-code framework with enterprise naming, tagging, and deployment patterns used across ML and application workloads.
- Built a data integration platform (Python, SQL, DuckDB) with a FastAPI/React console, and manage the PostGIS databases and AWS (Lambda, S3, SageMaker) architecture behind it.
- Built and deployed GIS applications (ArcGIS Enterprise, Experience Builder, Dashboards, Utility Network) for infrastructure monitoring and disaster response — spatial web portals used in real time during Hurricane Milton for structural impacts, outages, and crew deployment.

Associate Geospatial Analyst — Florida Power & Light - Contract

Sep 2024 – Mar 2025

- Led QA/QC of LiDAR-derived power grid data for infrastructure resilience work, including pole hardening.
- Managed PostgreSQL/PostGIS databases, optimized Cloud-Optimized GeoTIFFs, and automated QA workflows in FME, ArcPy, and Model Builder to streamline data storage and validation.
- Built a FastAPI sidecar for heavier geoprocessing and format conversion.

Graduate Research Assistant — Florida Atlantic University

Jan 2023 – Aug 2024

- Led flood risk research using FEMA hydrologic/hydraulic methods, resulting in a peer-reviewed publication on flood hazard exposure and social vulnerability — including environmental justice disparities in disaster impact.
- Built an automated ArcGIS Pro geoprocessing pipeline, cutting processing time by 50%.
- Built a Python geocoding tool processing 3 million addresses, replacing a paid service — cut turnaround from 3 months to 1 week and saved \$20K in licensing.

Geomatics Engineer / GIS Intern — Freelance & ADMC Engineering, Nepal

Dec 2020 – Dec 2022

- Led survey teams for municipal road, control, and construction surveys, including GNSS landmark and UAV topographic/transmission-line surveys.
- Designed municipal geodatabases (land use, soil, cadastral, resource maps) with topology and attribute rules.
- Served as GIS expert for Nepal's Survey Department land-use hand-over program, training 40+ officials in spatial analysis and database management.

EDUCATION

MSc, Geosciences — GIS Concentration — Florida Atlantic University · 2023–2024 · GPA 4.0/4.0

BSc, Geomatics Engineering — Tribhuvan University, Nepal · 2016–2021 · GPA 3.64/4.0

CERTIFICATIONS & TRAINING

The Rise of the AI Data Engineer Bootcamp (2026) · FAA Part 107 — Remote Pilot Certificate · Machine Learning Specialization — DeepLearning.AI · Deep Learning Specialization — DeepLearning.AI · GIS Specialization — UC Davis · GIS Mapping & Spatial Analysis — U. Toronto · Data Science with R — Johns Hopkins · Remote Sensing Image Acquisition & Analysis — UNSW Canberra · Introduction to Databases — Meta

NASA ARSET Certifications (12): Applied remote sensing training (2022–2024) across climate risk assessment & projection, disaster & building-infrastructure risk modeling, wildfire and watershed impacts, ecosystem services, air quality (geostationary satellites & environmental justice), water quality (inland lakes & SeaDAS), soil moisture/drought, flood monitoring & modeling, and urban heat vulnerability.

Esri Training Certifications (8): GIS Basics, Cartography, Spatial Data Science, Imagery Analysis, Spatial Analysis, GIS for Climate Action, GIS/BIM integration for AEC, and deep-learning object classification in ArcGIS Pro.

PUBLICATIONS

Dhungana, B.; Liu, W. *Urban–Rural Exposure to Flood Hazard and Social Vulnerability in the Conterminous United States*. ISPRS Int. J. Geo-Inf. 2024, 13, 339.

Co-author *PyWMP* — a Python watershed modeling and planning framework. Florida Atlantic University (manuscript in review).

PROJECTS

SurgeExposure: A cloud-native pipeline overlaying NOAA storm-surge and flood-inundation data against Overture Maps building footprints, scoring per-asset flood exposure across 8 coastal regions — with a map and API. (*Python, DuckDB Spatial, FastAPI, PostGIS, AWS*) [Demo](#) | [Repo](#)

fflood-nep: A reproducible flood-response pipeline for Nepal's August 2026 Rasuwa/Bhote Koshi flash flood — runs Sentinel-1 SAR change detection for flood extent, scores building/road/facility exposure by municipality, and tracks a separate InSAR pipeline for landslide/avalanche source detection. Built and shipped inside the event's first week. (*Python, Sentinel-1 SAR, STAC, GDAL, InSAR*) [Demo](#) | [Repo](#)

wildfire-copilot: A LangGraph agent for utility wildfire risk — combines a trained XGBoost susceptibility model with live NASA/NWS hazard feeds and RAG over real wildfire-mitigation filings to answer plain-English risk questions and prioritize inspections. (*LangGraph, XGBoost, RAG, FastAPI*) [Demo](#) | [Repo](#)

geomlint: A CI-first data-quality linter for vector geospatial files — catches invalid geometry, CRS confusion, and topology errors that mainstream data-observability tools skip. No GDAL required; pip install and it runs anywhere pip does. (*Python, Shapely, PyProj, PyPI, CLI*) [Docs](#) | [Repo](#)

Geospatial Data Copilot: Ask a plain-English question about San Francisco's utility infrastructure — a local LLM agent (LangGraph ReAct, Ollama) translates it to spatial SQL, runs it, and answers with a summary and interactive map. No API keys. (*LangGraph, Ollama, DuckDB Spatial, Chroma*) [Demo](#) | [Repo](#)

HazardMCP: An MCP server unifying FEMA, USGS, NOAA, NWS, and NASA hazard data behind a single AI-agent interface — one-call location risk summaries and map rendering, compatible with Claude Desktop. (*Python, FastMCP, Folium, Docker*) [Repo](#)

Road-Centerline: A published Python package that extracts road centerlines from polygon geometries — worldwide, in any input CRS, in any format GeoPandas can read. CLI and Python API, computed via medial-axis skeletonization. (*Python, GeoPandas, pygeoops, PyProj, PyPI*) [Repo](#)

3DEP LiDAR Terrain Builder: Generates DEM, DSM, hillshade, slope, aspect, and contour products from USGS 3DEP lidar. Draw an area of interest in the browser; it generates a configured local script — no data ever uploads. (*PDAL, GDAL, RichDEM, AWS EPT/S3*) [Demo](#) | [Repo](#)

GeoCoder: Converts a spreadsheet of US street addresses into geocoded points and building footprint polygons using free Nominatim and Overpass APIs — as a no-install web app or a scriptable Jupyter notebook. (*GeoPandas, Nominatim / Overpass, Static Web App*) [Demo](#) | [Repo](#)

GTFS Transit Platform: Data-engineering and analytics platform for GTFS transit feeds — headways, stop coverage, and spacing anomalies, explored in an interactive map app. [Demo](#) | [Repo](#)

3d-export: 3D-style land cover export from Sentinel-2 tiles, clipped to any country and automated via GitHub Actions — a modernized take on the rayshader R workflow. [Demo](#) | [Repo](#)

RomanNepaliAI: Converts romanized Nepali to Devanagari live, client-side, plus a CLI that batch-translates SRT subtitles through pluggable backends. [Demo](#) | [Repo](#)

Conversational AI Assistant: LangChain RAG chatbot with a history-aware retriever for multi-turn, context-grounded conversation, citing sources. Runs locally via Ollama. [Repo](#)

compound-risk-pbc: Client-side visualization tool for compound risk scoring — interactive charts over probability-based combination models. [Demo](#) | [Repo](#)

fv-gwr: Analysis code behind a peer-reviewed study on urban–rural flood hazard exposure and social vulnerability across the U.S. [Repo](#)

Air-Pollution-in-Florida: Dashboard visualizing daily mean PM2.5 across Florida with an automated update pipeline. [Repo](#)

CURRENTLY BUILDING

groundtruth: An agentic tool-execution layer that wraps every LLM-driven GIS operation in deterministic CRS resolution, geometry validation, and topology checks.

vecinity: A vector index that spatially partitions as it searches, instead of bolting H3 cell IDs onto a generic ANN index as a metadata filter.

fv-gmgwr: GPU-accelerated Multi-scale Geographically Weighted Regression (MGWR) in PyTorch, extending the modeling approach behind my peer-reviewed flood-exposure paper (Dhungana & Liu, 2024).